

Module 1: Intelligent Agents & PEAS

Definitions of AI, Turing Test, Rationality, PEAS Framework & 5 Agent Types

Module I: Introduction to Artificial Intelligence & Intelligent Agents

1. Foundations & 4 Definitions of AI (Human vs. Rational, Thought vs. Behavior)

3. Historical Evolution & Eras of AI (Dartmouth 1956 to Deep Learning)

5. The PEAS Framework (Performance, Environment, Actuators, Sensors)

7. Environment Taxonomies (7 Orthogonal Dimensions & Classifications)

9. Model-Based Reflex Agents (Handling Partial Observability & State Updating)

11. Utility-Based Agents (Trade-Off Optimization & Expected Utility Theory)

13. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)
2. The Turing Test, Total Turing Test & Chinese Room Argument

4. Concept of Rationality & Mathematical Performance Measures

6. Comprehensive PEAS Analysis across 5 Diverse Real-World Domains

8. Simple Reflex Agents (Condition-Action Rules & Internal Architectures)

10. Goal-Based Agents (Search, Planning & Future State Simulation)

12. Learning Agents (Critic, Learning Element, Performance Element, Generator)

Topic 1: Foundations & The 4 Quadrants of AI Definitions

Artificial Intelligence (AI) is the study and design of intelligent computational entities capable of perceiving their environment, reasoning about state, making optimal decisions, and acting autonomously to maximize success.

Dimension	Human-Centric Approach	Rationalist (Normative Ideal) Approach
Thought Processes & Reasoning	Systems that Think Like Humans: • Focuses on Cognitive Science & Cognitive Modeling. • Validated by comparing internal neural activation traces and decision response times with human psychological experiments (GPS - General Problem Solver).	Systems that Think Rationally: • Focuses on Formal Logic & Laws of Thought. • Codifies strict deductive syllogisms (e.g., Aristotle: "Socrates is a man; all men are mortal; therefore Socrates is mortal").
Observable Behavior & Action	Systems that Act Like Humans: • Operationalized by The Turing Test (1950). • Requires Natural Language Processing, Knowledge Representation, Automated Reasoning, and Machine Learning.	Systems that Act Rationally (Modern AI Standard): • Focuses on Intelligent Rational Agents. • An agent acts rationally if it selects actions that maximize the expected value of its performance measure, given its perceptual history.

Topic 2: The Turing Test & The Chinese Room Philosophical Argument

1. The Standard Turing Test (Alan Turing, 1950):

A human interrogator communicates via text terminal with two hidden entities: another human and a machine. If the interrogator cannot reliably tell which is the machine after 5 minutes of open interrogation, the machine is said to have demonstrated intelligence.

Total Turing Test: Augments the text test with video/physical interactions, requiring Computer Vision and Robotics.

2. John Searle's Chinese Room Argument (1980):

A human who understands zero Chinese sits inside a sealed room with a rule book (program) that maps incoming Chinese character slips to output slips. To an outside observer, the room answers Chinese queries perfectly.

Core Philosophical Takeaway: Syntax vs. Semantics

Searle proved that **syntactic symbol manipulation** is fundamentally insufficient for **semantic understanding and intentionality** ("Strong AI"). Machines simulate understanding without truly understanding.

Topic 4 & 5: Concept of Rationality & The PEAS Framework

A **Rational Agent** is one that does the right thing. Formally, for each possible percept sequence, a rational agent selects an action that is expected to maximize its performance measure, based on the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rationality $\neq$ Omniscience

- **Omniscience:** An omniscient agent knows the *actual* outcome of its actions and can act accordingly (impossible in reality due to incomplete information and stochasticity).
- **Rationality:** Maximizes *expected* success based on available percepts and knowledge. Rationality encourages **information gathering and exploration**.

The PEAS Formulation Matrix across 5 Standard AI Systems:

Agent Type	Performance Measure (<i>P</i>)	Environment (<i>E</i>)	Actuators (<i>A</i>)	Sensors (<i>S</i>)
1. Autonomous Taxi Driver	Safe transit, fast route, legal driving, passenger comfort, maximized profits.	Roads, traffic, pedestrians, weather, traffic lights, police.	Steering wheel, accelerator, brake, signals, horn, dashboard display.	LiDAR, RADAR, cameras, sonar, GPS, speedometer, accelerometer, engine sensors.
2. Medical Diagnostic System	Healthy patient, minimized healthcare costs, zero false negatives in critical diagnoses.	Patient, hospital clinical staff, insurance databases.	Display screen for diagnoses, recommended prescriptions, test orders.	Keyboard input of symptoms, lab test analyzers, vital sign monitors, patient EHR.
3. Automated Vacuum Cleaner	Cleanliness score, minimal battery consumption, minimal operating time, zero damage to furniture.	Room floors, carpets, furniture, obstacles, pets, stairs.	Drive wheels, vacuum suction motor, rotating brush, dust ejection valve.	Bumper contact switches, infrared cliff sensors, optical floor sensor, dust detector.
4. Part-Picking Industrial Robot	Percentage of parts correctly sorted into bins, speed (parts per minute), zero damage.	Conveyor belt, raw parts bins, assembly trays.	Jointed mechanical robotic arm, pneumatic gripper hand.	High-resolution RGB-D camera, joint angle encoders, tactile pressure sensors.
5. High-Frequency Trading Bot	Maximized portfolio return on investment (ROI), minimized Value-at-Risk (VaR), Sharpe ratio.	Global financial exchanges (NASDAQ, NYSE, crypto liquidity pools).	Automated FIX protocol buy/sell order submissions, cancellation packets.	Real-time tick-by-tick market data feed, financial news APIs, order books.

Topic 7: Environment Taxonomies (7 Orthogonal Dimensions)

Environment Property	Description & Characterization	Representative Domain Examples
1. Fully vs. Partially Observable	Whether the agent's sensors give complete access to the full state of the environment at all times.	• <i>Fully</i>: Chess, Crossword puzzle. • <i>Partially</i>: Poker (hidden cards), Automated Driving.
2. Single-Agent vs. Multi-Agent	Whether other entities in the environment are treated as agents with their own objective functions.	• <i>Single</i>: Solitaire, Sudoku. • <i>Multi</i>: Chess (Competitive), Traffic (Cooperative).
3. Deterministic vs. Stochastic	Whether the next state is completely determined by the current state and the agent's action.	• <i>Deterministic</i>: Chess. • <i>Stochastic</i>: Backgammon (dice), Weather-dependent systems.
4. Episodic vs. Sequential	In episodic environments, current decisions do not affect future episodes (no memory needed).	• <i>Episodic</i>: Defect classification on conveyor. • <i>Sequential</i>: Chess, Navigation.
5. Static vs. Dynamic vs. Semidynamic	Whether the environment changes while the agent is deliberating. Semidynamic: environment stays fixed but score decreases over time.	• <i>Static</i>: Crossword. • <i>Dynamic</i>: Taxi driving. • <i>Semidynamic</i>: Timed Chess.
6. Discrete vs. Continuous	Whether the state space, time, percepts, and actions are finite discrete values or continuous real numbers.	• <i>Discrete</i>: Chess (64 squares). • <i>Continuous</i>: Self-driving cars (speed, angle).
7. Known vs. Unknown	Whether the physics/rules of the environment are fully known to the agent in advance.	• <i>Known</i>: Solitaire. • <i>Unknown</i>: Exploring a new video game or planet.

Topics 8 – 12: The 5 Fundamental Agent Architectures

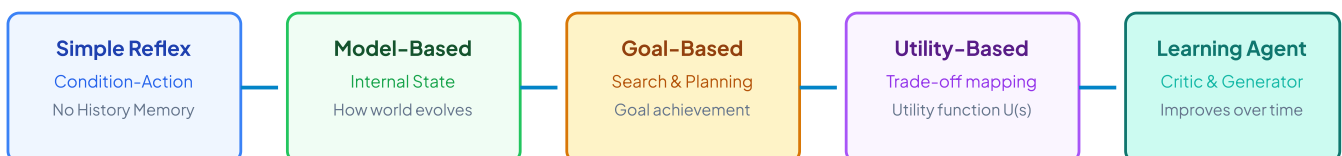


Figure 1.1: Progression of Autonomous Agent Architectures from Simple Reflex to Learning Agents

1. The Learning Agent Sub-Components:

Critic: Evaluates the agent's behavior against an external performance standard and provides reward/penalty feedback.

Learning Element: Responsible for making improvements based on feedback from the critic.

Performance Element: Responsible for selecting external actions (equivalent to the entire non-learning agent).

Problem Generator: Suggests novel exploratory actions that lead to new and informative experiences, rather than sticking only to known suboptimal paths.

Q1. Define an Intelligent Agent. Differentiate between Omniscience and Rationality with an example. (8 Marks)

An **Intelligent Agent** is anything that perceives its environment through sensors and acts upon that environment through actuators.

- **Rationality:** Evaluated based on the expected performance given the percept sequence observed so far and built-in knowledge. It represents optimal decision-making under uncertainty.
- **Omniscience:** An omniscient agent knows the exact actual outcome of its actions and makes zero errors (impossible due to incomplete state visibility and random environmental events).

Example: A pedestrian crosses the street after carefully looking left and right (Rational). If a meteor strikes them from the sky, the action was still rational despite the tragic outcome, whereas an omniscient agent would have known about the meteor.

Q2. Provide a complete PEAS description for an Automated Medical Diagnosis System and analyze its environment across all 7 dimensions. (10 Marks)**PEAS Description:**

- **P:** Patient health recovery, minimal diagnostic cost, zero misdiagnoses of critical illnesses.
- **E:** Patient physiology, hospital staff, clinical lab test systems, electronic health records.
- **A:** Diagnostic reports, prescription medication orders, clinical test recommendations.
- **S:** Symptom inputs, patient vital sign telemetry, lab test results, imaging scans.

Environment Classification:

1. *Partially Observable:* Internal physiological processes cannot be fully sensed directly.
2. *Multi-Agent:* Interacts with human doctors, nurses, and patients.
3. *Stochastic:* Patient biological responses to medication vary probabilistically.
4. *Sequential:* Previous treatment choices alter patient physiology and future treatment efficacy.
5. *Dynamic:* Patient condition can deteriorate while the system deliberates.
6. *Continuous:* Vital signs, lab values, and time are continuous variables.
7. *Known:* Medical physics and pharmacology rules are largely established.